

OPEN RESEARCH DATA MANAGEMENT PLAN (DMP)

Center for High Entropy Alloy Catalysis (CHEAC)
- from April 2026

Term	Description
UCPH	University of Copenhagen, host institution of CHEAC and provider of institutional infrastructure.
ERDA	ERDA (Electronic Research Data Archive, University of Copenhagen). Long-term storage and public archival platform used for frozen research data archives.
ASIMOV	Internal CHEAC data management system implementing a structured Project → Measurement hierarchy for organising raw data, metadata, and scripts prior to archival. Used for online data analysis and structured sharability of data among peers.
DFT	Density Functional Theory. Quantum mechanical method used to compute electronic structure properties in materials simulations.
GPAW	Density functional theory (DFT) software used for electronic structure simulations.
VASP	Vienna Ab initio Simulation Package, DFT software used for electronic structure simulations.
Niflheim	High-performance computing (HPC) cluster at Technical University of Denmark (DTU) used for large-scale computational simulations.
SCIENCE HPC Center	High-performance computing infrastructure at the University of Copenhagen used for computational research.
SFTP	Secure File Transfer Protocol. Encrypted file transfer method used to retrieve experimental data from external facilities as well as some internal instruments.
CC BY 4.0	Creative Commons Attribution 4.0 licence under which project data is released for reuse.
README file	Plain text documentation file providing contextual information about the contents of a directory, including description, parameters, provenance, and relevant usage information.
Duplicati	Duplicati (https://github.com/duplicati/duplicati) is an open-source backup software that provides automated, encrypted, and integrity-verified backups to remote storage such as ERDA.

SUMMARY

CHEAC generates two primary categories of research data: computational simulations, mainly DFT calculations performed with GPAW or VASP, and experimental measurements acquired on internal- or external instruments (UCPH / collaborators), and at external large research facilities.

Computational data includes input files, submission scripts, raw text output containing full parameter specifications and software versions, and additional output files such as .gpw, .traj and .db.

Experimental data includes raw instrument output in native or text-based formats, together with acquisition parameters, calibration information, and analysis scripts. Data obtained at external facilities is retrieved via SFTP and incorporated into the same internal structure as locally acquired data.

Typical data volume is up to 150 GB per person per year for both computational and experimental work, depending on technique.

The research does not involve personal or sensitive data. All research data is intended for open release.

DATA STRUCTURE AND INTERNAL STORAGE STANDARD

All research data, computational and experimental, must be organised according to a consistent hierarchical structure to ensure traceability, reproducibility, and reuse.

- Data is organised at two logical levels: the research project level and the level of individual datasets arising from specific experiments or simulations.
- Each research project directory must contain a top-level README file describing the scientific objective, responsible researchers, collaborators, relevant publications, and an overview of included datasets. Within this directory, each experiment or simulation is stored in its own subdirectory.
- Each dataset directory must contain the complete raw data exactly as produced by the instrument or simulation software. It must also contain a README file documenting the instrumentation or software version, acquisition or input parameters, date of execution, responsible researcher, and a concise description of purpose. Any preprocessing, analysis, or figure-generation scripts must be stored within the same directory. Where relevant, references to related experiments or simulations must be included.
- Raw data must *never* be overwritten. Processed data must be stored separately and clearly labeled to distinguish it from original output. Scripts used to generate published figures must remain in the same structured directory as the data they operate on.
- Experimental data initially resides on laboratory acquisition computers and must be transferred to long-term storage as soon as practical, and all laboratory computers must implement automatic encrypted backups to ERDA using Duplicati or an equivalent solution that provides comparable encryption standards, data integrity guarantees, and failure resilience.
- Data acquired at external facilities must be retrieved via SFTP and incorporated into the same structured research project directory. Computational data generated on Niflheim or the SCIENCE HPC Center may remain on those systems during active work, as both provide backup. Upon project completion or manuscript submission, this data must be incorporated into the structured research project directory and deposited as a frozen archive on ERDA.

All research data stored within CHEAC must satisfy the **C-L-E-A-R** standard:

C – Complete: All raw data must be preserved in full. No original files may be overwritten or deleted once incorporated into a research project directory.

L – Labeled: Every dataset must contain a README file documenting software or instrument version, acquisition or input parameters, date, responsible researcher, and scientific purpose.

E – Executable: All analysis and figure-generation scripts used in publications must be stored alongside the data they operate on. Workflows must be reproducible from the archived directory.

A – Archived: Data must be transferred from temporary storage (laboratory computers, HPC scratch space, or external facilities) into structured project storage and subsequently deposited on ERDA at manuscript submission.

R – Reusable: Data must be organised and documented such that an external researcher can understand and reuse it without additional clarification. Processed data must be clearly distinguished from raw data.

Implementation example: ASIMOV implements this structure directly. Exporting a research project from ASIMOV produces a fully structured archive containing documentation, raw data, metadata, and scripts. This archive can be deposited on ERDA without modification.

IMPLEMENTATION OF F-A-I-R DATA PRINCIPLES

F – Findable: Upon manuscript submission, the complete project directory associated with the publication is exported and deposited as a public frozen archive on ERDA. The archive link and identifier are referenced in the manuscript and on the group webpage. Metadata recorded in the archive includes author names, title, description, date, and DOI where applicable. Because the internal project structure is preserved in the archive, contextual information remains directly accessible.

A – Accessible: All data underlying submitted manuscripts is made publicly available at submission. ERDA guarantees availability of frozen archives for at least ten years and supports datasets up to at least 10 TB. Data is accessible via direct download as compressed archives. Both computational and experimental datasets are released together with scripts and documentation required to reproduce the results.

I – Interoperable: Reproducibility is ensured by preserving raw outputs, parameter specifications, software versions, and analysis workflows within the same structured directory. Where possible, plain text formats are used to ensure long-term readability independent of specific software environments. Methodological details are described in the associated manuscript, while operational details, parameter settings, and scripts are contained directly within the archive.

R – Reusable: Data is released under a Creative Commons Attribution (CC BY 4.0) licence without embargo. Documentation within each project and measurement directory enables external researchers to understand experimental setups, simulation conditions, and processing workflows without requiring additional clarification. Because raw data and scripts are preserved together, future researchers can reproduce figures, modify analyses, or reuse datasets in new contexts.

DATA SECURITY AND LONG-TERM PRESERVATION

Computational data is generated on Niflheim and the SCIENCE HPC Center, both of which provide backup during active work. Transfer into the structured project directories is required at manuscript submission, in connection with archival on ERDA. Experimental data is temporarily stored on laboratory computers before transfer to ASIMOV and/or ERDA. Data retrieved from external facilities via SFTP follows the same workflow.

All services require authenticated access with two-factor authentication and are backed up daily. ERDA provides replicated storage and remote disaster recovery. ASIMOV is backed up and mirrored remotely. Long-term preservation is ensured through ERDA frozen archives.